

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

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Code of Conduct:

I Harish Kumar / 192521340 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. Answer: Number of segments = 150 MB (150,000,000 bytes) ÷ 1500 bytes = 100,000 segments. Since each segment is encapsulated into one frame, the number of frames = 100,000.

Data Link Layer overhead = $100,000 \times 40 \text{ bytes} = 4,000,000 \text{ bytes (4 MB)}$. Total data transmitted (payload + overhead) = $100,000 \times 1540 \text{ bytes} = 154,000,000 \text{ bytes} = 1,232,000,000 \text{ bits}$. Transmission time = $1,232,000,000 \text{ bits} \div 50,000,000 \text{ bps} \approx 24.64 \text{ seconds}$. The Transport Layer ensures reliability through segmentation, sequencing, acknowledgments and retransmission of lost segments, while the Data Link Layer ensures reliability over each physical link through framing, error detection (CRC) and, where used, automatic repeat request (ARQ).

2. Answer: Number of transmission windows = $48 \text{ frames} \div 6 \text{ frames per window} = 8 \text{ windows}$. If one frame is lost in every window, total retransmissions = $1 \times 8 = 8 \text{ retransmissions}$. Flow control (sliding window) improves efficiency by allowing the sender to transmit multiple frames before waiting for an acknowledgment, matching the sender's rate to the receiver's processing/buffer capacity, reducing idle time compared to stop-and-wait, and preventing buffer overflow at the receiver.

3. Answer: Usable data per fragment = $\text{MTU} - \text{IP header} = 1500 - 20 = 1480 \text{ bytes}$. Number of fragments = $\text{ceil}(12,000 \div 1480) = 9 \text{ fragments}$ (8 fragments of 1480 bytes and 1 fragment of 160 bytes). Total header overhead = $9 \times 20 \text{ bytes} = 180 \text{ bytes}$. The Network Layer ensures successful delivery of fragmented packets using the identification field (to group fragments belonging to the same packet), the fragment offset field (to reassemble them in the correct order) and the more-fragments flag, with reassembly performed at the destination host.

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5. Answer: Checkpoints occur every 60 MB — at 60, 120, 180, 240, 300 and 360 MB — so 6 checkpoints are created before the failure at 390 MB. Data transmitted after the last checkpoint (360 MB) = $390 - 360 = 30 \text{ MB}$, which must be retransmitted after recovery. Synchronization checkpoints in the Session Layer are important because they allow a failed transfer to resume from the last checkpoint instead of restarting from the beginning, saving time and bandwidth and supporting orderly dialog control and recovery.

6. Answer: Compressed size = $900 \text{ MB} \times (1 - 0.40) = 540 \text{ MB}$. Encrypted size = $540 \text{ MB} \times 1.05 = 567 \text{ MB}$. Total reduction compared to the original file = $(900 - 567) \div 900 \times 100 = 37\%$. The Presentation Layer improves efficiency by compressing data to reduce the volume transmitted, and improves security by encrypting data so it cannot be read if intercepted; it also handles data translation/formatting so information is understood consistently by sender and receiver.

7. Answer: Number of requests = $3 \text{ GB (3,000 MB)} \div 3 \text{ MB} = 1,000 \text{ requests}$. Total data = $3 \text{ GB} = 3,000,000,000 \text{ bytes} = 24,000,000,000 \text{ bits}$. Transmission time = $24,000,000,000 \text{ bits} \div 120,000,000 \text{ bps} = 200 \text{ seconds}$. The Application Layer provides reliable services through protocols such as FTP/HTTP that structure requests, handle authentication and error handling, and rely on the underlying Transport Layer (e.g., TCP) for guaranteed, ordered delivery.

8. Answer: Total end-to-end delay = processing delays ($4 + 3 + 5 + 2 + 1 = 15 \text{ ms}$) + propagation delay (18 ms) + transmission delay (12 ms) = 45 ms. Each OSI layer contributes

processing delay as it encapsulates/decapsulates and examines headers (Application, Transport, Network, Data Link, Physical); propagation delay depends on the physical medium and distance; transmission delay depends on packet size and link bandwidth. Together these delays determine the total latency experienced by the packet.

9. Answer: Payload per frame = $1600 - 80 = 1520$ bytes. Number of frames = $80 \text{ MB} (80,000,000 \text{ bytes}) \div 1520 \text{ bytes} \approx 52,632$ frames. Total overhead = $52,632 \times 80 \text{ bytes} \approx 4,210,560 \text{ bytes} (\approx 4.21 \text{ MB})$. Transmission efficiency = $\text{useful data} \div \text{total data transmitted} \times 100 = 1520 \div 1600 \times 100 = 95\%$. Protocol overhead reduces the effective (goodput) throughput available for useful data; a higher overhead relative to frame size lowers efficiency, especially for smaller frames, and affects overall network performance.

10. Answer: Compressed file size = $240 \text{ MB} \times (1 - 0.30) = 168 \text{ MB}$. Number of segments = $168 \text{ MB} (168,000,000 \text{ bytes}) \div 1400 \text{ bytes} = 120,000$ segments. Total Data Link Layer overhead = $120,000 \times 60 \text{ bytes} = 7,200,000 \text{ bytes} (7.2 \text{ MB})$. Final data transmitted across the Physical Layer = $168,000,000 + 7,200,000 = 175,200,000 \text{ bytes} (175.2 \text{ MB})$. Multiple OSI layers work together: the Presentation Layer compresses (and can encrypt) the data for efficiency and security, the Transport Layer segments the data and ensures reliable end-to-end delivery, the Network Layer routes the segments to the destination, the Data Link Layer frames the data and adds error control for each hop, and the Physical Layer transmits the resulting bits — together ensuring the patient record is delivered securely and reliably.